CMSC 2621
Lab Exercise 1

Due: See the assignment folder for the due date.

Overview:

Embark on a thrilling treasure hunt adventure by solving puzzles and challenges using pointers in C++. Along the way, you'll encounter hidden traps, secret codes, and ancient relics. Your goal is to find the treasure while mastering pointer concepts!

Requirements:

1. Download the PointerTreasureHunt.cpp file that's linked in the D2L assignment. Follow the instructions in the comments and update only the locations in the code that have a "CHANGE_HERE"
2. Submit the source code and a document of a screenshot in a zip file to the D2L assignment. (Please see the instructions in the end of this document)

Resources:

Use the resources below:

https://cplusplus.com/doc/tutorial/pointers/

https://learn.microsoft.com/en-us/cpp/cpp/pointers-cpp

Submission Requirements:

You'll be submitting a zip file that contains the files below:

1. The source code file
2. A Word document containing screenshots of your source code and the full results of executing your program.

After creating a zip file containing all the required files, submit the zip file to the assignment folder on D2L.

Screenshot Requirements:

1. The screenshot(s) need to be clear enough to read the source code.
2. If the running results are too long to be captured, use multiple screenshots.
3. The running result(s) need to be captured in the screenshot(s) completely. If the running results are not complete, i.e. just partial results are captured in the screenshot(s), you may also receive a points deduction, depending on how complete the results in the screenshots are.

Assignment Grading:

1. If the program fails to be compiled using a compiler that supports C++ 20, a maximum grade of 40% will be given.
2. If any requirements are not met, for each unfulfillment, a corresponding penalty will be given.
3. Completeness of the lab assignment will also be considered.

Expected Output:

Any reference/address output will be different on each execution:

```
Treasure Location: 10
Treasure Location using mapPointer: 10
Map Reference Code: 0109FE2C
Key Locations on the Map:
1 2 3 4 5
Traps Avoided: 20
Crossing the Trap Field:
0 0 1 0 1
Before swap: Code A = 1234, Code B = 5678
After swap: Code A = 5678, Code B = 1234
Master Key Value: 30
```